

Exponents and Positive Powers of 10: Summary

Summary

(a) **Exponential Notation**

Exponential notation is shorthand for repeated multiplication of the same quantity. A number written in exponential notation has a base and an exponent. The exponent indicates the number of times to multiply the base. For example, the product $8 \cdot 8 \cdot 8$ is written as 8^3 in exponential notation. The number 8 is the base and the number 3 is the exponent.

(b) **Positive Powers of 10**

A **positive power of 10** is a power of 10 in which the exponent is a positive integer: $\{1, 2, 3, 4, \dots\}$. Below are the first four positive powers of 10.

$$\begin{aligned}10^1 &= 10 \\10^2 &= 100 \\10^3 &= 1,000 \\10^4 &= 10,000\end{aligned}$$

In general, if n is a positive integer, then the value of 10^n is the number 1 followed by n zeros.

(c) **Multiplying a Number by a Positive Power of 10**

To multiply a number by a positive power of 10, count the number of zeros in the power of 10, then move the decimal point to the **right** that many places.

(d) **Dividing a Number by a Positive Power of 10**

To divide a number by a positive power of 10, count the number of zeros in the power of 10, then move the decimal point to the **left** that many places.